

Derivatives of Inverse Trig Functions

Numberbender | WORKSHEET



Name: _____ Date: _____ Score: _____

Find dy/dx for each function. Show all work.

Calculus 1 Worksheet #21 Derivatives of Inverse Trig Functions and Implicit Differentiation

EXAMPLES:

1. The derivative of $y = \cos^{-1}(5x)$ is $-\frac{1}{\sqrt{1-x^2}}(d\Box) = -\frac{1}{\sqrt{1-(5x)^2}}(5) = \boxed{-\frac{5}{\sqrt{1-25x^2}}}$
2. The derivative of $y = \tan^{-1}(4x^3)$ is $\frac{1}{1+\Box^2}(d\Box) = \frac{1}{1+(4x^3)^2}(12x^2) = \boxed{\frac{12x^2}{1+16x^6}}$

EXAMPLE FOR IMPLICIT DIFFERENTIATION :

1. Find dy/dx for $y^2 - 4x^3 + xy = 23$
3. $2y\left(\frac{dy}{dx}\right) - 12x^2\left(\frac{dx}{dx}\right) + \left[\left(\frac{dx}{dx}\right)y + x\left(\frac{dy}{dx}\right)\right] = 0$
- $$2y\left(\frac{dy}{dx}\right) - 12x^2 + y + x\left(\frac{dy}{dx}\right) = 0 \Rightarrow 2y\left(\frac{dy}{dx}\right) + x\left(\frac{dy}{dx}\right) = 12x^2 - y \Rightarrow$$
- $$(2y + x)\left(\frac{dy}{dx}\right) = 12x^2 - y \Rightarrow \frac{dy}{dx} = \frac{12x^2 - y}{2y + x}$$

For the following fourteen problems, find $\frac{dy}{dx}$.

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|----------------------------|--|--|
| 1. $y = \sin^{-1}(2x)$ | 2. $y = \tan^{-1}(3x)$ | 3. $y = \sec^{-1}(e^{2x})$ |
| 4. $y = \sin^{-1}\sqrt{x}$ | 5. $y = \sin^{-1}\left(\frac{x}{3}\right)$ | 6. $y = \cos^{-1}(2x + 1)$ |
| 7. $y = \sec^{-1}(x^7)$ | 8. $y = \csc^{-1}(e^x)$ | 9. $y = \sin^{-1}\left(\frac{1}{x}\right)$ |
| 10. $y = e^x(\sec^{-1}x)$ | 11. $y = x^2(\sin^{-1}x)^3$ | |
| 12. $y^2 - 8y + x^2 = 5$ | 13. $x^2y^3 = 25$ | 14. $y^3 - 6y + x^2 = 0$ |



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Answer key — for instructor use only.

Answers:

1. $\frac{2}{\sqrt{1-4x^2}}$	2. $\frac{3}{1+9x^2}$	3. $\frac{2}{\sqrt{e^{4x}-1}}$	4. $\frac{1}{2\sqrt{x-x^2}}$
5. $\frac{1}{\sqrt{9-x^2}}$	6. $\frac{-2}{\sqrt{1-(2x+1)^2}}$	7. $\frac{7}{ x \sqrt{x^{14}-1}}$	8. $\frac{-1}{\sqrt{e^{2x}-1}}$
9. $\frac{-1}{ x \sqrt{x^2-1}}$	10. $\frac{e^x}{ x \sqrt{x^2-1}} + e^x \sec^{-1} x$	11. $\frac{3x^2(\sin^{-1} x)^2}{\sqrt{1-x^2}} + 2x(\sin^{-1} x)^3$	12. $\frac{-x}{y-4}$
13. $-\frac{2y}{3x}$	14. $-\frac{2x}{3(y^2-2)}$		

